

3.1.6	Alkanes	
	<i>Fractional distillation of crude oil</i>	know that alkanes are saturated hydrocarbons know that petroleum is a mixture consisting mainly of alkane hydrocarbons understand that different components (fractions) of this mixture can be drawn off at different levels in a fractionating column because of the temperature gradient
	<i>Modification of alkanes by cracking</i>	understand that cracking involves the breaking of C–C bonds in alkanes know that thermal cracking takes place at high pressure and high temperature and produces a high percentage of alkenes (mechanism not required) know that catalytic cracking takes place at a slight pressure, high temperature and in the presence of a zeolite catalyst and is used mainly to produce motor fuels and aromatic hydrocarbons (mechanism not required) understand the economic reasons for the cracking of alkanes (e.g. ethene used for poly(ethene); conversion of heavy fractions into higher value products)
	<i>Combustion of alkanes</i>	know that alkanes are used as fuels and understand that their combustion can be complete or incomplete and that the internal combustion engine produces a number of pollutants (e.g. NO_x, CO and unburned hydrocarbons) know that these pollutants can be removed using catalytic converters know that combustion of hydrocarbons containing sulfur leads to sulfur dioxide that causes air pollution and understand how sulfur dioxide can be removed from flue gases using calcium oxide know that the combustion of fossil fuels (including alkanes) results in the release of carbon dioxide into the atmosphere know that carbon dioxide, methane and water vapour are referred to as greenhouse gases and that these gases may contribute to global warming